Fujiao Ji

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RESEARCH INTERESTS

Web Security; LLM/VLM Security; LLM Distillation

EDUCATION

University of Tennessee, Knoxville Aug. 2022 – Present
Ph.D. in Computer Science TN, USA
Advisor: Dr. Doowon Kim

Shandong University of Science and Technology Aug. 2018 – Jun. 2021
M.S. in Computer Software and Engineering Qingdao, China
Advisor: Prof. Zhongying Zhao

EXPERIENCE

University of Tennessee, Knoxville Aug. 2022 - Present
Graduate Research Assistant, Graduate Teaching Assistant TN, USA
Phishing website investigation.

Shandong University of Science and Technology Aug. 2018 - Jun. 2021
Graduate Research Assistant, Graduate Teaching Assistant Qingdao, China
Heterogeneous graph representation learning.

Baidu Map Jun. 2021 - Jan. 2022
Research Internship, directed by Dr.Yibo Sun and Dr. Lei Shao Beijing, China
Named entity recognition for Chinese address data.

SELECTED PROJECTS

Red-Team Evaluation of LLM-Generated Phishing Websites Jun. 2025 - Present

- Designed *PhishGen*, an adversarial evaluation framework spanning 14 commercial, open-source, and uncensored LLMs, 7 multimodal prompt configurations, and 12 prompt-rewriting strategies; generated and analyzed 30,192 phishing website samples across four major brands.
- Built an end-to-end testing pipeline (Python, Playwright, HTML/CSS analysis, automated visual metrics, and browser-based credential submission) to measure safeguard bypass, visual fidelity, frontend rendering, backend execution, and credential persistence; found 96.2% of generated pages rendered successfully while only 38.7% completed the full credential-harvesting workflow.
- Identified novel model- and generator-level security risks, including near-perfect DOM imitation, systematic CSS styling omissions, and a strong textual-over-visual brand mimicry bias; translated these findings into actionable signals for detecting and mitigating LLM-generated phishing attacks.

Security Evaluation of LLM-Based Phishing Detection Jan. 2025 - Present

- Built *PhishBench*, a multimodal benchmark of 9,567 real-world phishing websites (APWG eCX, Apr.–Oct. 2025) and 400 manually verified benign sites across 200 brands; ran the controlled comparison of 4 LLMs (GPT, Gemini, Qwen, Llama) against 2 DL baselines over 4 input configurations.
- Found Gemini delivers the best trade-off (89.7% detection at a 2.0% FPR) and GPT the most conservative (73.7% at 0.5%), while open-source LLMs incurred up to 48.0% FPR (Qwen) and DL baselines, though far faster (1.1 s/sample), missed most phishing pages (FNR up to 78.1%).
- Conducted failure analysis revealing that GPT appears anchored to a few salient signals, Gemini responds broadly across Normal-looking features, Qwen shows sparse and potentially unstable cue use, and Llama appears less visually accompanied and more reliant on textual-structural signals.

Adversarial Evaluation of Visual Anti-Phishing Systems Aug. 2022 - Dec. 2024

- Conducted a large-scale adversarial evaluation of visual anti-phishing models using 451K real-world phishing websites collected through APWG.
- Identified and characterized three real-world evasion strategies used by attackers to bypass visual similarity-based detection mechanisms.
- Analyzed detection failures and identified three recurring evasion strategies–pipeline exploitation, benign-logo mimicry, and simple visual manipulation–revealing a 20.7% average performance degradation from curated benchmarks to real-world phishing data.
- Designed an adversarial robustness benchmark of 7,223 screenshots spanning 14 visible and five perturbation-based logo manipulations; uncovered transferable vulnerabilities and derived defenses based on adversarial augmentation, OCR–vision integration, multi-cue ensembles, and image preprocessing.

Chinese Address Parsing Project Jun. 2021 - Jan. 2022

- Extracted and structured address data in the format of province, city, district, town, and point of interest.
- Recognized named entities through a biaffine attention mechanism based on the pre-trained model ERNIE 1.0 under the framework of PaddlePaddle and improved performance via post-processing.
- Evaluated the performance of point of interest chunks, where the F1 score is 81.25% for 1,000 real-world data from Baidu Map and 80.41% for 2,985 public data from Chinese Address Corpus.

Heterogeneous Networks Analysis Sept. 2018 - Jun. 2019

- Conducted literature reviews on the work related to heterogeneous networks.
- Made a comparative study on heterogeneous networks and classified them into four categories according to topological and attribute information.

SELECTED PUBLICATIONS

- **Ji, Fujiao**, and Doowon Kim. 'How Can We Effectively Use LLMs for Phishing Detection?: Evaluating the Effectiveness of Large Language Model-based Phishing Detection Models.' arXiv preprint arXiv:2511.09606 (2025).
- **Ji, Fujiao**, Kiho Lee, Hyungjoon Koo, Wenhao You, Euijin Choo, Hyoungshick Kim, and Doowon Kim. 'Evaluating the effectiveness and robustness of visual similarity-based phishing detection models.' In 34th USENIX Security Symposium, pp. 3201-3220. 2025.
- Lim, Kyungchan, Kiho Lee, **Fujiao Ji**, Yonghwi Kwon, Hyoungshick Kim, and Doowon Kim. 'What's in Phishers: A Longitudinal Study of Security Configurations in Phishing Websites and Kits.' In Proceedings of the ACM on Web Conference, pp. 957-968. 2025.
- Elgedawy, Ran, Porter Dosch, John Sadik, Senjuti Dutta, Anuj Gautam, Konstantinos Georgiou, Farzin Gholamrezae, **Ji, Fujiao**, et al. 'Occasionally secure: A comparative analysis of code generation assistants.' arXiv preprint arXiv:2402.00689 (2024).
- **Ji, Fujiao**, Zhongying Zhao, Hui Zhou, Heng Chi, and Chao Li. 'A comparative study on heterogeneous information network embeddings.' Journal of Intelligent & Fuzzy Systems 39, no. 3 (2020): 3463-3473.

SKILLS

- **Languages & Framework**: Python, Pytorch, Tensorflow, PaddlePaddle
- **Skills**: CTF, Agent, LLM Distillation, LLM Finetuning, Claude Code, Codex, Linux, Tmux, vLLM, Hugging Face, Azure, LaTex, Pandas, Graph

PROFESSIONAL ACTIVITIES

- PC Member for USENIX 2027
- Peer Reviewer for Pattern Recognition
- Artifact Evaluation Committee for NDSS 2027, USENIX 2026, IEEE S&P 2026